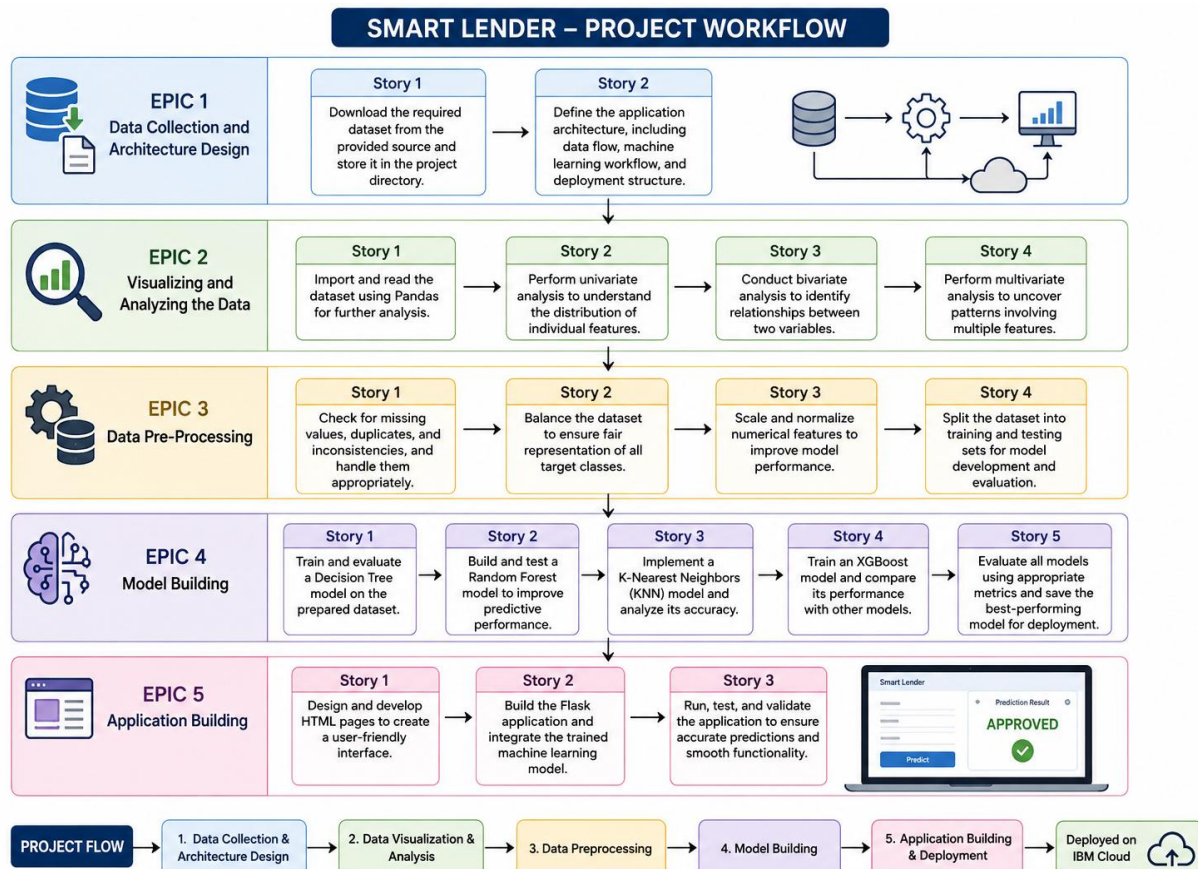


Project Workflow (Epics and User Stories)

Date	28 June 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender

Project Workflow (Epics and User Stories)

The Smart Lender project follows a structured machine learning development lifecycle divided into five major epics. Each epic contains user stories that define the activities required to successfully design, develop, evaluate, and deploy the loan approval prediction system.



Project Workflow (Epics and User Stories)

Epic 1: Data Collection and System Architecture

Story 1: Obtain the Loan Approval Prediction dataset from the specified source and organize it within the project directory.

Story 2: Design the overall system architecture, including data flow, machine learning pipeline, Flask application structure, database interactions (if applicable), and IBM Cloud deployment workflow.

Epic 2: Data Exploration and Visualization

Story 1: Import the dataset using the Pandas library and inspect its structure, data types, and summary statistics.

Story 2: Perform univariate analysis to understand the distribution of individual features such as applicant income, loan amount, and credit history.

Story 3: Conduct bivariate analysis to identify relationships between independent variables and the loan approval status.

Story 4: Perform multivariate analysis using correlation matrices and visualization techniques to uncover hidden patterns and feature interactions.

Epic 3: Data Preprocessing

Story 1: Detect and handle missing values, duplicate records, and inconsistent data entries.

Story 2: Encode categorical variables and balance the dataset to improve model fairness and prediction accuracy.

Story 3: Normalize or standardize numerical features to enhance machine learning model performance.

Story 4: Split the processed dataset into training and testing sets for model development and evaluation.

Epic 4: Machine Learning Model Development

Story 1: Train and evaluate a Decision Tree classifier.

Story 2: Build and evaluate a Random Forest classifier.

Story 3: Train and analyze a K-Nearest Neighbors (KNN) classifier.

Story 4: Train an XGBoost classifier and compare its performance with the other models.

Story 5: Evaluate all models using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. Select and save the best-performing model for deployment.

Project Workflow (Epics and User Stories)

Epic 5: Web Application Development and Deployment

Story 1: Design responsive HTML pages to provide a user-friendly interface for loan application submission.

Story 2: Develop the Flask application and integrate the trained XGBoost model for real-time prediction.

Story 3: Test and validate the application to ensure accurate loan approval predictions and reliable functionality.

Story 4: Deploy the Smart Lender application on IBM Cloud, verify successful deployment, and perform end-to-end testing in the cloud environment.